Project Objectives

1. **Develop** an intelligent, multi-agent Infrastructure-as-Code (IaC) validation framework that supports multiple IaC technologies such as CloudFormation, Terraform, AWS CDK, Kubernetes, and AWS Serverless Application Model (SAM) for smart-manufacturing cloud environments.

2. **Improve** the deployment success of LLM-generated or LLM-repaired Infrastructure-as-Code templates by incorporating automated validation, deployment verification, and intelligent feedback mechanisms.

3. **Enhance** the detection and mitigation of security misconfigurations, including insecure IAM policies, network configurations, and cloud resource settings, using LLM-assisted validation and automated remediation.

4. **Detect** and report configuration drift by correlating Infrastructure-as-Code definitions with runtime cloud telemetry, deployment logs, and resource states to ensure infrastructure consistency.

5. **Provide** explainable and cost-aware recommendations that improve the security, reliability, maintainability, and operational efficiency of Infrastructure-as-Code deployments while reducing manual validation effort.